MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 2SWF

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper, conductor, in accordance with ASTM B 286, size 22-7, tin coated. Maximum diameter 0.033 inch.

Second - Polyethylene insulation, natural, type II, class L, grade 4 in accordance with L-P-390. Nominal thickness 0.010 inch. Colored insulation, one black and one white conductor for each pair.

Third - Clear polyamide jacket on each conductor in accordance with ASTM D 4066. Nominal thickness 0.003 inch.

Fourth - Two conductors, one black and one white, cabled together with nominal lay of 2.5 inches to form each pair.

Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.

Sixth - Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.

Seventh - Shield insulation of one polyester tape, type G in accordance with MIL-I-631 plus a jacket of clear polyamide in accordance with ASTM D 4066. Nominal thickness of 0.003 inch. Standard identification code applied by method 2.

- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

Eighth - The required number of pairs (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. Nonfibrous fillers shall be employed.

FSC 6145

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Ninth - Synthetic rubber compound filled binder tape applied helically with overlap. (Surface of tape shall be such that cable jacket will adhere to it.)

Tenth - Arctic tape polychloroprene jacket (black). (See table I for thickness.) Cable surface marking required.

TABLE I. Details.

Type and size	Number of pairs	Cable jacket thickness minimum (inch)	Overall diameter	
			Minimum (inch)	Maximum (inch)
2SWF-3	3	0.100	0.600	0.625
2SWF-4	4	.090	.600	.625
2SWF-7	7	.110	.780	.815
2SWF-24	24	.125	1.190	1.250

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	3000
Conductor to shield	1500
Shield to shield	100
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage 1000 lbf/in^2, in 6 hours, in^3, (test each shipping length)	Zero
Capacitance	
Mutual capacitance - at 1 MHz, pF/ft, maximum	30
Capacitance unbalance - percent, maximum	8
Characteristic impedance - at 1 MHz, ohms	75 ± 5
Group B:	
Attenuation - at 3 MHz, db/100 feet, maximum	3
Cold working (minus 54 ± 2°C)	No damage
Drip - at 75 ± 1°C	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - lbf/in^2, minimum	1400
Elongation - percent, minimum	300
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	300
Set - inch, maximum	3/8

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - percent of unaged, minimum..	60
Permanence of printing (jacket) - cycles, minimum.....	250
Cable filler removability.....	No failure
Shield-conformance to material coverage and construction	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (95 ± 3°C).....	No failure
Cold working (minus 54 ± 2°C).....	No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal).

Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0885-43)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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